

worksheet is presented. But it does not offer much difficulty as the application of Eqs. (3.26)–(3.29) requires no other built-in function.

3.4.2 Detect Outliers and Stragglers

The initial application of the precision model developed in ISO 5725 standards was for commercial purposes. At present, many regulatory bodies are using it as a prerequisite to accept any analytical method for official control. For example, when a new ISO standard operating procedure is developed, it is mandatory to conduct an interlaboratory study and publish the precision values obtained in an informative annex. As explained, the other application of this standard algorithm is the computation of intermediate precision parameters. It is less promoted, although it is fully presented in standard ISO 5725 and especially useful for Chapters 5 and 6.

The official exploitation of reproducibility has many practical consequences. Unfortunately, exceptionally distant data values may be present and alter the results, and a large part of the ISO standards is devoted to the detection of non-normal values, called outliers and stragglers. This issue is less important when the computation is applied to intermediate precision. When considering Figure 3.2, three kinds of outliers may be suspected, which are displayed in Figure 3.6. A full collection of statistical tests is proposed to verify several types of rejection hypotheses. They are applied at two values of α error risk:

- If $\alpha > 5\%$ (or $\alpha > 2.5\%$ in some guidelines) the data is a straggler.
- If $1\% > \alpha > 5\%$ (or $0.5\% > \alpha > 2.5\%$) it is an outlier.

The decision rule varies depending on the α -value. Given the cost of recruiting a laboratory and the fact that interlaboratory study parameters are only publishable

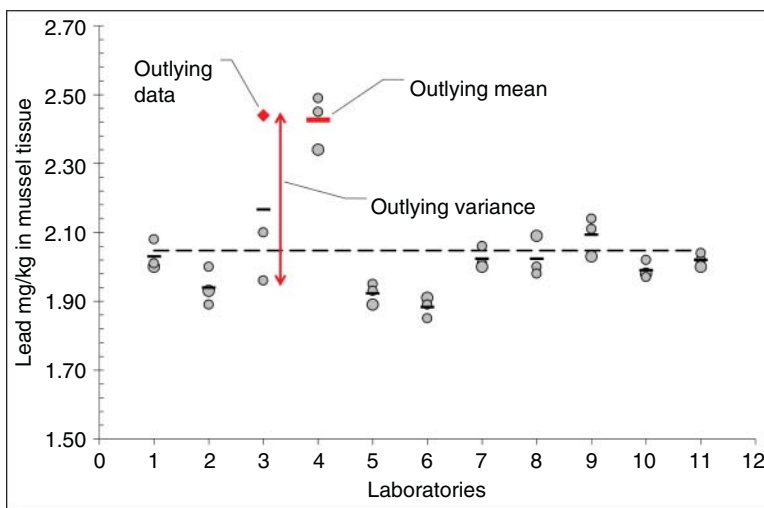


Figure 3.6 Diverse types of outliers in an interlaboratory study.

when at least eight laboratories obtain acceptable results, it was interesting to consider the possible rejection of one outlying replicate to keep a laboratory. Statistical literature is very abundant on the question of outlier significance testing. Several dozen authors have described tests for rejecting outliers from a dataset.

A limited set of rejection tests was selected by the group of experts in charge of preparing the ISO 5725 standard. Because many outlier rejection tests exist but may not always identify the same data as an outlier, it seemed important to the experts to clearly put a well-identified list in the standard. The complete operating procedure for each particular test is not presented here but is described in the standard for the interested reader. Anyway, the specific task of data polishing is left to the statistician in charge of the study. Apart from Dixon's test, when a laboratory is identified as an outlier, all its data are usually eliminated. The diverse outliers are listed here with their corresponding official tests.

Hypothesis	Overestimated	Official test
One outlying variance	Repeatability	Cochran's
One or two outlying averages	Reproducibility	Grubbs' single or double
One outlying replicate in a lab	Lab variance	Dixon's

Because outlier rejection is an overly sensitive subject for analysts, many criticisms were issued once the proposed tests were published. Some of these objections are relevant. For example, with Dixon's test, it is possible to declare that one data out of four is an outlier and the distribution is non-Gaussian. This consequence seems strange when it may take more than 1000 measurements to prove that a variable is distributed according to a normal distribution. Depending on the sequence of application of the tests, it is not always the same laboratories that are eliminated, resulting in different estimates of precision.

To reduce this risk of inconsistency, it was also decided to define a standardized testing sequence for eliminating data. The best known is the ISO protocol that defines, on the one hand, a sequence of application of the tests and, on the other hand, an acceptance criterion for the whole study. The criterion consists of calculating the ratio of excluded laboratories to the total number of participants. When more than 2/9 laboratories have been eliminated after the following sequence of tests, the method is not validated, i.e. it is not publishable as a standard or used for commercial purposes. The ISO protocol is as follows:

- (a) Cochran's test at 5% risk.
- (b) Grubbs' test to cell replicates having a suspect variance; then eliminate outliers.
- (c) Go back to (a) if only one replicate is eliminated, otherwise, go to (d).
- (d) Cochran's test at 1% risk to all variances and eliminate cells with outliers.
- (e) Go back to (a) if only one cell is eliminated, otherwise, go to (f).
- (f) Apply simple Grubbs' test at 1% risk to all remaining cell means.
- (g) If a mean is eliminated, return to (f), otherwise, go to (h).

- (h) Apply double Grubbs test at 1% risk to all remaining cell means.
- (i) Calculate the proportion of eliminated laboratories and verify that it $< 2/9$.

The alternative protocol promoted by the International Union of Pure and Applied Chemistry (IUPAC) is similar to the ISO protocol but does not include the Grubbs' test for the elimination of outliers in a laboratory. This sequence of rejection tests is extraordinarily complex and, even when automated, requires manual decision-making. We have presented in some detail this very touchy topic of outliers because it is a repeated concern for analysts. It shows that even a consensus solution among expert statisticians does not result in a very satisfactory answer. The conclusion can be drawn that much attention must be paid to outlier rejection and statistical testing must be applied cautiously.

To illustrate the consequence of the rejection of non-normal data, outliers were eliminated from the LEAD dataset: the complete subset of laboratory 04 and one extreme measure of laboratory 03, as illustrated in Figure 3.7. When doing this operation, the experimental design is no longer balanced. Modification of the initial Resource E worksheet is necessary to manage the new dataset. Modified formulas are presented in Section 3.3 and coded in the new worksheet named Resource F, at the end of the chapter. It is mainly the formula used to compute the between-laboratory variance, which is modified in cell B25 as reported in Eq. (3.29).

The consequence of the outlier elimination is striking. The reproducibility standard deviation s_R is divided by 30, coming from 0.1642 down to 0.00540 mg/kg, as well as the limit of reproducibility R . Now, the largest acceptable difference between the two laboratories is no longer 0.47 mg/kg but only 0.07 mg/kg. It is easy to understand the economic consequences on international trade when the analytical method is used for the official control of Pb in foods. The control laboratories must be more efficient, and consumer health must be better preserved.

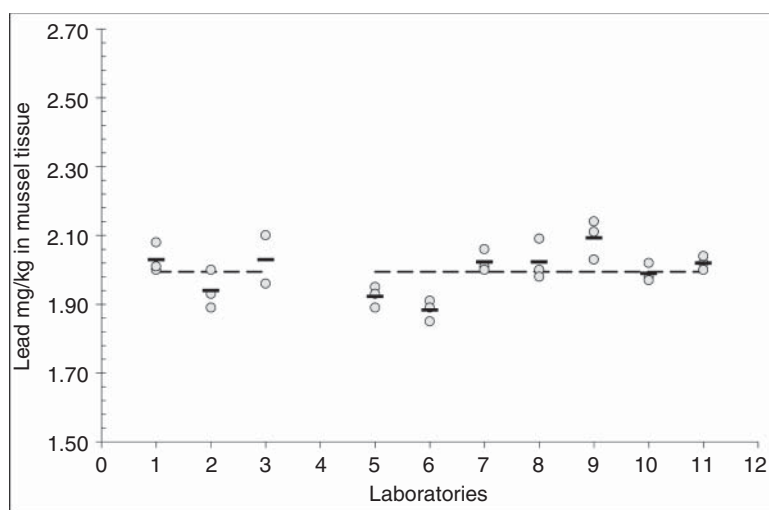


Figure 3.7 LEAD – interlaboratory study after outlier deletion.

Resource F Precision parameters for an unbalanced design (Excel).

	A	B	C	D	E	F
1	Resource F: Precision parameters for an unbalanced design					
2	Lead mg/kg in mussel tissue					
3	Laboratories	Rep. 1	Rep. 2	Rep. 3	SSi	ni²
4	Lab 01	2.08	2.00	2.01	0.003800	9
5	Lab 02	2.00	1.93	1.89	0.006200	9
6	Lab 03	2.10		1.96	0.009800	4
7	Lab 04					
8	Lab 05	1.95	1.89	1.93	0.001867	9
9	Lab 06	1.85	1.91	1.89	0.001867	9
10	Lab 07	2.01	2.00	2.06	0.002067	9
11	Lab 08	2.00	2.09	1.98	0.006867	9
12	Lab 09	2.11	2.03	2.14	0.006467	9
13	Lab 10	2.02	1.98	1.97	0.001400	9
14	Lab 11	2.02	2.00	2.04	0.000800	9
15	Main parameters					
16	Number of laboratories (I)	10	=COUNT(B4:B14)			
17	Number of measurements (IJ)	29	=COUNT(B4:D14)			
18	Number of replicates (J)	2.9	=B17/B16			
19	Sum of squared numbers	85	=SUM(F4:F14)			
20	Correct measurements number	26.1	=B17-(B19/B17)			
21	SSW	0.04113	=SUM(E4:E14)			
22	SSt	0.14492	=DEVSQ(B4:D14)			
23	SSB	0.10378	=B22-B21			
24	Repeatability variance (s²r)	0.00216	=B21/(B17-B16)			
25	Temporary between lab variance	0.00323	=(B16-1)*((B23/(B16-1))-B24)/B20			
26	Between laboratory variance (s²L)	0.0032337	=IF(B25<0;0;B25)			
27	Reproducibility variance (s²R)	0.00540	=B24+B26			
28	Precision					
29	Grand mean	1.994	=AVERAGE(B4:D14)			
30	Repeatability std. dev.	0.0465	=SQRT(B24)			
31	Interlaboratory std. dev.	0.0569	=SQRT(B26)			
32	Reproducibility std. dev.	0.0735	=SQRT(B27)			
33	Relative std. dev. Reproducibility (RSDR)	3.68%	=B32/B29			
34	Grand mean variance (s²Z)					
35	Variance ratio A	1.49	=B26/B24			
36	Coefficient Q	0.468	=(B35+1)/(B18*B35+1)			
37	Grand mean variance (s²Z)	0.0015	=B27*SQRT(1/(B17*B36))			
38	Grand mean std. dev. (sZ)	0.0383	=SQRT(B37)			

3.4.2.1 Other Algorithms

As shortly explained, it is common to encounter several problems when eliminating outliers. For example, if an obvious outlier (for the analyst) is not detected, too many laboratories are excluded, the study cannot be published, the precision is underestimated, etc. Finally, the estimates of repeatability and reproducibility are not satisfactory. For all these reasons, the ISO 5725 standard includes a set of *robust estimators* that have the advantage of being weakly affected by the presence of outlying data. The median is the typical example of a robust parameter for statisticians. In this context, the meaning of the word robustness is slightly different from that used in evaluating the analytical method's robustness. Finally, when robust estimators are used, it is pointless to eliminate suspect data.

The principle of the robust algorithms proposed by ISO is to iteratively transform the data by using robust location and dispersion estimators. Thus, it is possible

to calculate robust versions of the precision parameters, namely $\overline{\overline{Z}}$, s_r , and s_R . The application of these algorithms to interlaboratory studies is not developed here. It is mainly of interest to professional interlaboratory study organizers. For the details, it is possible to refer to Part 5 of the ISO 5725 standard or to the brief published by the Analytical Methods Committee of the Royal Society of Chemistry [6, 12, 13].

However, an example of a robust parameter applied to proficiency testing schemes is described in Section 4.3.4 and Resource G; it is officially recognized as Algorithm A. It gives an insight into the principle of this more recent statistical method. This evolution is pointed out here because it marked the introduction of new statistical methods based on iterative calculation in standards. Experience shows that it is interesting to compare the precision values obtained by the classic calculation method, before and after eliminating outliers, with those of the robust algorithm. In general, obtained values are in-between, which probably better reflects reality. However, the standardized robust methods are not fully satisfactory because they opaquely replace *problematic* data with other values closer to the rest of the dataset in such a manner that data traceability is lost.

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